MSMS 408 : Practical 09

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Question

Generate a random sample from $N(0,1)$ by the rejection method.

Theory

Note that, if $Z \sim N(0,1)$, $|Z|$ has the probability density function as follows.

$$f(x) = \frac{2}{\sqrt{2\pi}} e^{-x^2/2},\ 0 < x < \infty. \tag{1}$$

We start by generating from the preceding density function by using the rejection method with g being the exponential density function with mean 1, that is

$$g(x) = e^{-x},\ 0 < x < \infty.$$

Now,

$$\frac{f(x)}{g(x)} = \sqrt{\frac{2}{\pi}} e^{x - x^2/2}$$

and the maximum value of $\dfrac{f(x)}{g(x)}$ occurs at the value of x that maximizes $x - \dfrac{x^2}{2}$.

$$\begin{aligned} \frac{d}{dx}\left(x - \frac{x^2}{2}\right) &= 0 \\ \Rightarrow 1 - x &= 0 \\ \Rightarrow x &= 1. \end{aligned}$$

So we take

$$c = \text{Max}\frac{f(x)}{g(x)} = \frac{f(1)}{g(1)} = \sqrt{\frac{2e}{\pi}}.$$

Then

$$\begin{aligned} \frac{f(x)}{c \cdot g(x)} &= \exp\left\{x - \frac{x^2}{2} - \frac{1}{2}\right\} \\ &= \exp\left\{-\frac{(x-1)^2}{2}\right\}. \end{aligned}$$

Thus we can generate the absolute value of a standard normal random variable as follows:

I. Generate Y, an exponential random variable with rate 1.

II. Generate a uniform random number U.

III. If $U \leq \exp\left\{-\frac{(Y-1)^2}{2}\right\}$, set $X = Y$, otherwise return to step I.

Once we have simulated a random variable X having density function as in (1) — and such a random variable is thus distributed as the absolute value of a standard normal — we can then obtain a standard normal Z by letting Z be equally likely to be either X or $-X$.

R Program

```
n <- 10^5
```

```
y <- rexp(n, rate = 1)
```

```
u1 <- runif(n)
```

```
good <- u1 <= exp(-(y - 1)^2 / 2)
accepted <- y[good]
```

```
u2 <- runif(n)
```

```
z <- ifelse(u2 <= 1/2, accepted, -accepted)
```

Thus we have a sample of size 76339 from $N(0,1)$.

The **empirical acceptance rate** was 0.76339 whereas the **theoretical acceptance rate** is 0.7601735.

Let us have a visualization of the rejection technique.

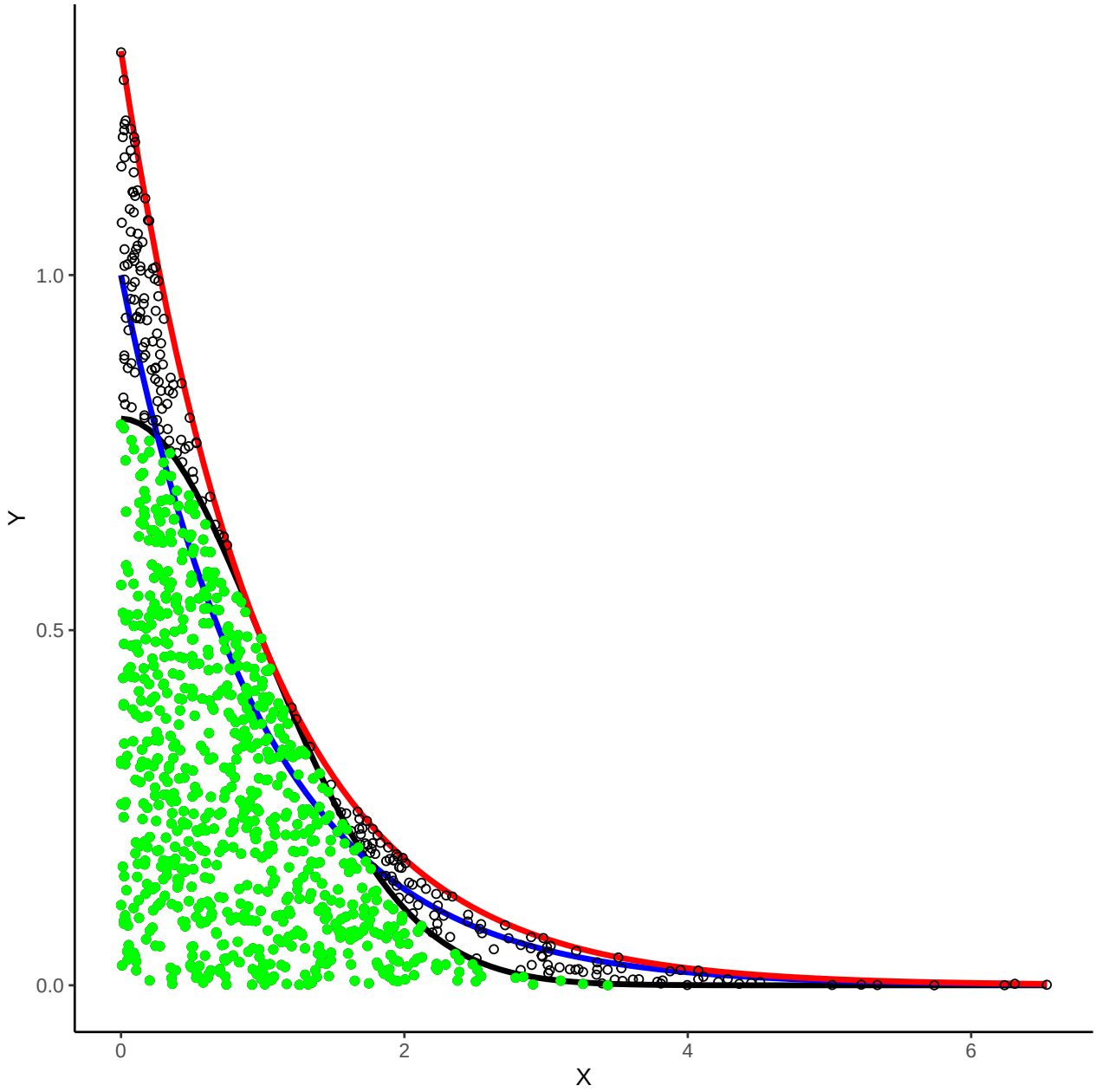
```
f_abs_Z <- function(x) 2 * dnorm(x)
```

```
g1 <- function(x) sqrt((2 * exp(1)) / pi) * dexp(x)
```

```
X <- rexp(1000); Y <- runif(1000, min = 0, max = g1(X))
```

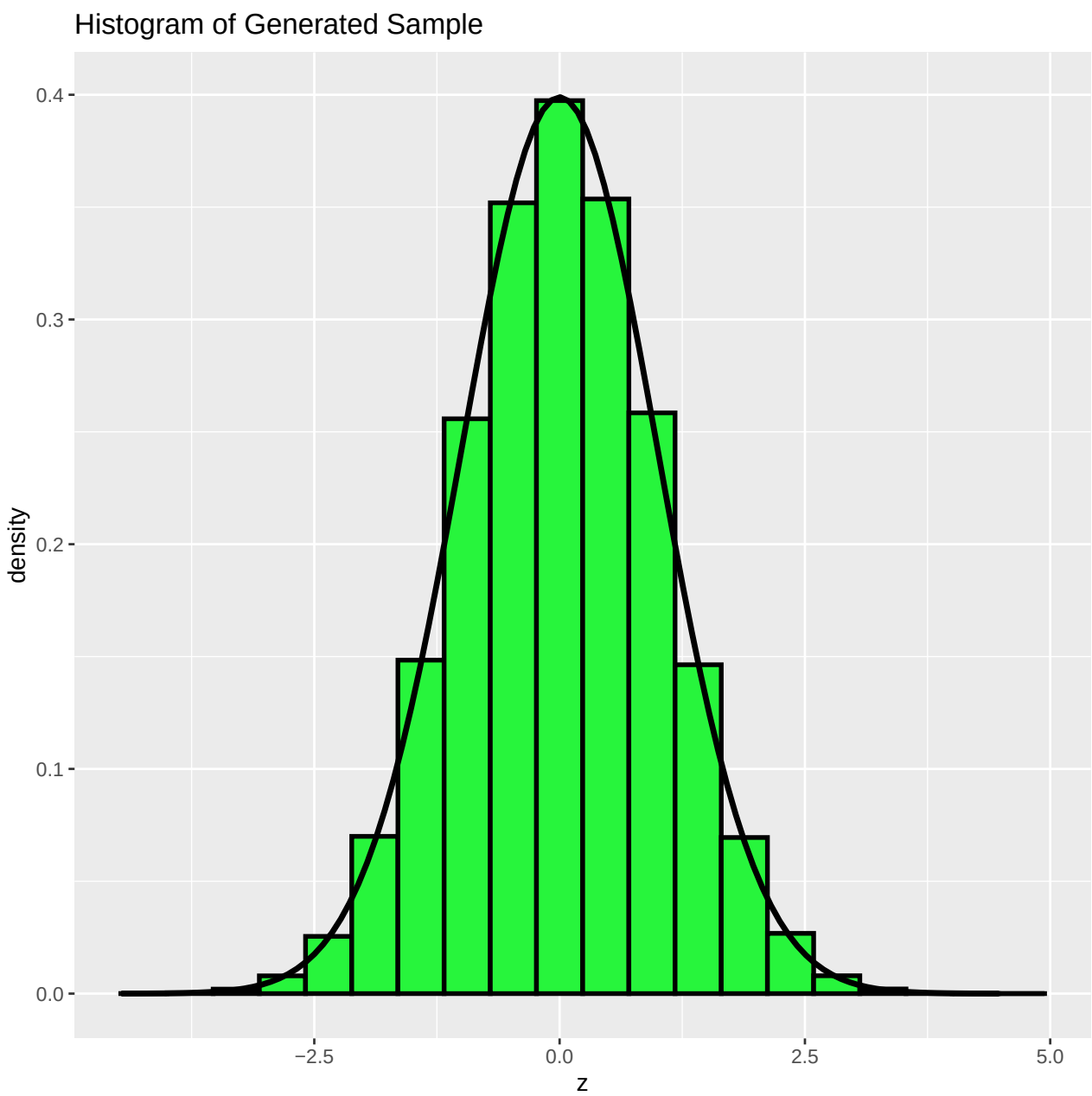
```
good <- Y <= f_abs_Z(X); accepted <- X[good]
```

```r
data.frame(x = c(0, 1)) %>%
  ggplot() +
  stat_function(fun = f_abs_Z, linewidth = 1.25) +
  theme_classic() +
  stat_function(fun = dexp, col = "blue", linewidth = 1.25) +
  stat_function(fun = g1, col = "red", linewidth = 1.25) +
  geom_point(data = data.frame(X, Y), aes(x = X, y = Y), shape = 1) +
  geom_point(data = data.frame(x = X[good], y = Y[good]), aes(x = x, y = y),
             col = "green")
```



Let us have a histogram of the generated sample.

```
data.frame(z) %>%
  ggplot(aes(x = z)) +
  geom_histogram(aes(y = after_stat(density)), bins = 20,
                 fill = "#27F53C", color = "black", linewidth = 1) +
  stat_function(fun = dnorm, linewidth = 1.25) +
  labs(title = "Histogram of Generated Sample")
```



✎ The PDF of a standard normal variate is well fit for the histogram of generated sample.